ZHAOHUI LI

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Research Interests

My research focuses on AI for education. My work bridges AI and education by translating educational goals and expert knowledge into task-specific computational tasks. I have developed algorithms, benchmark datasets, and systems that support teaching and learning, including automated assessment and feedback, instructional material evaluation, storybook generation, and special education. My career goal is to develop reliable, responsible AI frameworks that promote equitable, high-quality education worldwide.

My future research will continue pursuing interdisciplinary collaborations to develop AI applications that address specific educational needs. Building on these collaborations, I will develop a theoretical AI for education framework that translates pedagogical goals into computational tasks, along with an agent-based system to automate parts of task formulation and evaluation design.

Keywords: AI for Education, Human–AI Collaboration, Large Language Models, Natural Language Processing, Automated Assessment and Feedback, AI Literacy, Early Literacy, Special Education

Work Experience

- **College of AI, Cyber and Computing, The University of Texas at San Antonio** *Sep 2026 – Present*
  Postdoctoral Associate | *Mentor: Dr. Jinjun Xiong* San Antonio, TX
- **Institute for Artificial Intelligence and Data Science (IAD), Department of Computer Science and Engineering (CSE), University at Buffalo** *Aug 2024 – Aug 2026*
  Postdoctoral Associate | *Mentor: Dr. Jinjun Xiong* Buffalo, NY
  - AI Team Lead at the **National AI Institute for Exceptional Education (AI4ExceptionalEd)** (Aug 2024–present).
  - AI Team Lead at the **Center for Early Literacy and Responsible AI (CELaRAI)** (Aug 2024–present).

Education Experience

- **The Pennsylvania State University** *Aug 2019 – May 2024*
  Ph.D. in Computer Science — Natural Language Processing Lab State College, PA
  - **Advisor:** Dr. *Rebecca J. Passonneau*
  - **Dissertation:** *Automated and Semi-Automated Assessment of STEM Reasoning Questions.* **(Best Doctoral Dissertation Award)**
- **Nankai University** *Aug 2012 – Jun 2016*
  B.S. in Computer Science and Engineering Tianjin, China
  - **Advisors:** *Dr. Jun Xu* & *Dr. Jie Liu*
  - **Thesis:** BDA: Big Data Analysis as a Service (2k+ GitHub Stars)

Grants Experience

- **Organization for Autism Research (OAR) Research Grant ($50,000)** *Feb 2026 – Dec 2027*
  ***Co-PI**; Serve as the AI research lead. Parent's AI Coach for Parent-Implemented Communication Interventions for Children with ASD.*
- **IES NCSER — GAUGE (requested budget: $3,750,000)** *In submission*
  ***Co-PI**; From Guesswork to Judgment, an AI-Supported Simulation Platform for Training Preservice Teachers to Make Reliable, Data-Driven Accommodation Decisions. Transformative Research in Special Education (ALN 84.324T, FY2027); PI: Dr. Chunling Niu.*
- **Spencer Foundation — Large Research Grant on Education ($125,000–$500,000)** *TBD*
  ***Co-PI**; Serve as the AI research lead for the proposal AI-Automated Behavioral Support Plans for Young Children with Autism Spectrum Disorder. (Reject; resubmission in preparation).*
- **IES #R305C240046 — Center for Early Literacy and Responsible AI ($10,000,000)** *Sep 2024 – Aug 2029*
  ***Tech Lead**; Support Dr. Jinjun Xiong (AI Thrust Lead); developing responsible AI tools for early literacy and diverse learners.*
- **NSF #2229873 — AI Institute for Exceptional Education ($20,000,000)** *Jan 2023 – Dec 2027*
  ***Tech Lead**; Support Dr. Jinjun Xiong (Science Director); advancing AI solutions for challenges in special education.*
- **NSF IUSE #2236150 — Project CLASSIFIES ($300,000)** *Mar 2023 – Feb 2025*
  ***Contributor**; Developed NLP algorithms and helped write the technical proposal with Dr. Rebecca J. Passonneau (Co-PI).*
- **Penn State AI Challenge Grant — Nittany AI Challenge Award 2023 ($5,000)** *Jan 2023 – Jan 2024*
  ***Team Lead**; Developed an AI-driven Interactive E-Books System for Children.*

Selected Research Experience

- **PaiCoach: AI Assessment and Parent Coaching for Autism Home Interventions** *Nov 2024–Present*
  Research Lead | *Mentor: Prof. Jinjun Xiong*
  - Lead ASD-HI benchmark and PaiCoach AI model development as part of the AI4EE project, assessing parent–child interactions and generating feedback for specialist review [C7].
  - Submitted a dataset for benchmarking speech recognition for autistic children to LAK 2027. A related grant proposal to NSF Future CoRe is in preparation.
  - Secured a $50,000 OAR funding as CO-PI, extending this work to AI-supported parent coaching.
  - **Publications:** ASD-HI (*AIED 2025*) [C7]; PaiCoach (*Journal of Autism and Developmental Disorders*, 2026) [J1].

- **SciEval: Evaluating K–12 Science Instructional Materials** *July 2025–Present*
  Research Lead | *Mentor: Prof. Jinjun Xiong*
  Collaborators: Profs. Tingting Li and Peng He, Washington State University
  - Lead task formulation, rubric co-design, and model fine-tuning for evaluating science instructional materials; collaborate with education experts to validate LLM scores, reasoning, and suggestions [C2].
  - Collaborate on developing a platform to help teachers understand, evaluate, and adapt STEM instructional materials for classroom use. A proposal to NSF STEM K–12 is in preparation.
  - **Publications:** SciEval (*AIED 2026*) [C2]; Judging the Judges (*Computing Conference 2026*) [C4].

- **AI Reading Enhancer (AIRE): Decodable Story Generation and Evaluation** *July 2024–Present*
  Research Lead | *Mentor: Prof. Jinjun Xiong (Center Co-PI)*
  Collaborators: Prof. Tanya Christ, John Strong, and Laura Tortorelli (co-PIs)
  - Develop teacher-guided story-generation tools and lead AIRE development at the CELaRAI; co-design rubrics for phonics alignment, narrative coherence, and illustration quality with literacy experts.
  - **Publications/presentations:** StoryLab (*AIED 2025*) [C6]; AIRE (*AERA 2026 poster*) [C5].

- **Embodied AI: Reading Companions for Children's AI Literacy** *Nov 2024–Present*
  Research Lead | *Mentor: Prof. Jinjun Xiong*
  Collaborator: Prof. Shiyan Jiang, University of Pennsylvania
  - Implement pedagogically structured dialogue for an LLM-powered reading robot; evaluate dialogue policies using simulated child profiles and expert review [J2].
  - Collaborated on studies of bilingual conversational agents and dialogic reading, examining children's participation, learning, and engagement in AI-guided and parent-guided reading [C1].
  - **Publication:** Reading companion for children's AI literacy (*AIED 2026 Companion Proceedings*) [C1]; bilingual conversational reading (*Computers and Education: Artificial Intelligence*, 2025) [J2]; parent-led versus AI-guided reading (*BJET*, 2025) [J3].

- **LLM4Art: Expert-Guided Art Interpretation and Evaluation** *July 2025–Present*
  Research Lead | *Mentor: Prof. Jinjun Xiong*
  Collaborator: Prof. Kyunghee Pyun, Fashion Institute of Technology
  - Lead application and evaluation development, translating expert knowledge into prompts and rubrics for visual analysis, historical context, and cultural significance.
  - **Publication:** LLM4Art (*LAK 2026*) [C3].

- **Automated Formative Assessment of STEM Short Response** *Aug 2019–May 2024*
  Research Lead | *Mentor: Prof. Rebecca J. Passonneau, The Pennsylvania State University*
  Collaborators: Prof. Matthew D. Beckman, The Pennsylvania State University; Prof. Sadhana Puntambekar, University of Wisconsin–Madison
  - Developed SFRN and AsRRN to model relationships among questions, reference responses, and student answers; investigated demonstration examples for LLM assessment of science essays.
  - **Publications:** SFRN (*EMNLP 2021*) [C12]; AsRRN (*Findings of EMNLP 2023*) [C9]; science-essay assessment (*AIED 2025 workshop*) [C8].

- **STEM Assessment Datasets and Human Scoring Rubrics** *Aug 2019–May 2024*
  Research Lead | *Mentor: Prof. Rebecca J. Passonneau, The Pennsylvania State University*
  Collaborator: Prof. Matthew D. Beckman, The Pennsylvania State University

- Contributed to ISTUDIO and physics lab-report datasets, including rubric development and human-scoring procedures for automated assessment research.
- **Datasets:** ISTUDIO (2023) [R2]; rubric-based assessment of physics lab reports (2022) [R3].

- **Human–AI Collaboration: Selective Prediction and Deferral** *Aug 2019–May 2024*

Research Lead | *Mentor: Prof. Rebecca J. Passonneau, The Pennsylvania State University*
Collaborator: Prof. Sadhana Puntambekar, University of Wisconsin–Madison

- Developed methods for automated assessment models to answer or defer to a person, including joint optimization of assessment and deferral representations.
- **Publications:** Human deferral for short-answer grading (*AIED 2023*) [C10]; joint training for selective prediction (*arXiv preprint*, 2024) [R1].

Teaching Experience

- **Pennsylvania State University** *Jan 2020 – May 2020*

Teaching Assistant — Natural Language Processing (CSE597, graduate level) State College, PA

- Taught the large language models (LLMs) portion of the course.
- Graded assignments and supported student projects.

- **Pennsylvania State University** *Aug 2019 – Dec 2019*

Teaching Assistant — Artificial Intelligence (CMPSC442, undergraduate level) State College, PA

- Taught classes on Bayesian methods.
- Assisted with teaching, grading, and exams for over 100 students.

- **NSF Argumentation Workshop (PACE Lab)** *Feb 2022*

Invited Speaker Illinois State University

- Delivered a workshop talk on AI-assisted assessment and Large Language Models.

Service Experience

- **Proposal Reviewer:** NSF DRL Panelist on Career
- **Conference Reviewer:** ACL Rolling Review(ACL, EMNLP, NAACL), AAAI, AIED, LAK, SIGIR
- **Session Chair:** AIED 2025, 2026 (Language Learning & Story Generation, AI Agents for Education)
- **Program Committee Member:** AIED 2025 and 2026; AIED 2027 and LAK 2027
- **Student Volunteer:** AAAI 2019

Publications

Conference & Workshop Papers

[C1] Xiao, F., **Li, Z.**, Jiang, S., & Xiong, J. (2026). Design and Feasibility of an LLM-Powered Humanoid Robot as a Reading Companion to Support Children's AI Literacy. In *AIED 2026 Companion Proceedings* (pp. 498–504).

[C2] **Li, Z.**, He, P., Chen, Z., Liu, H., Wang, Z., Li, T., & Xiong, J. (2026). SciEval: A Benchmark for Automatic Evaluation of K–12 Science Instructional Materials. In *Proceedings of the International Conference on Artificial Intelligence in Education*. (AIED 2026)

[C3] **Li, Z.**, Li, N., Liu, D., Hu, Y., Xu, C., Chao, S., & Xiong, J. (2026). LLM4Art: Using Large Language Models for Art Interpretation. In *Companion Proceedings of the 16th International Conference on Learning Analytics & Knowledge (LAK26)* (pp. 246–248). *Zhaohui Li and Naiquan Li contributed equally and share first authorship.* Proceedings PDF.

[C4] He, P., **Li, Z.**, Wang, Z., Xiong, J., & Li, T. (2026). Judging the Judges: Human Validation of Multi-LLM Evaluation for High-Quality K–12 Science Instructional Materials. In *Proceedings of the Computing Conference*. (Computing Conference 2026)

[C5] **Li, Z.**, Zheng, Q., Liu, D., Christ, T. M., Strong, J. Z., Tortorelli, L. S., Wang, X. C., Alwan, A., Hakkani Tur, D., & Xiong, J. (2026, April). AIRE: AI Reading Enhancer for Personalized Decodable Texts in Early Literacy. Poster presented at the *American Educational Research Association Annual Meeting*, Los Angeles, CA. (AERA 2026)

[C6] **Li, Z.**; Xiao, F.; Lin, J.; Zou, X.; Zheng, Q.; Xiong, J. (2025, July). StoryLab: Empowering Personalized Learning for Children through Teacher-Guided Multimodal Story Generation. In *International Conference on Artificial Intelligence in Education* (pp. 285–292). Cham: Springer Nature Switzerland. (AIED 2025)

[C7] **Li, Z.**, Akemoglu, Y., Lyu, J., Zheng, Q., & Xiong, J. (2025, July). ASD-HI: A Parent-Child Interaction Dataset for Automated Assessment of Home Intervention. In *International Conference on Artificial Intelligence in Education* (pp. 48–62). Cham: Springer Nature Switzerland. (AIED 2025)

[C8] Sheikhi_Karizaki, M., **Li, Z.**, & Passonneau, R. J. (2025, July). Criteria for Demonstration Examples for LLM Assessment of Science Essays. In *Workshop on Epistemics and Decision-Making in AI-Supported Education, Annual Conference on Artificial Intelligence in Education*. (AIED 2025)

[C9] **Li, Z.**, Lloyd, S. E., Beckman, M. D., & Passonneau, R. J. (2023, December). Answer-state recurrent relational network (AsRRN) for constructed response assessment and feedback grouping. In *Findings of the Association for Computational Linguistics: EMNLP 2023* (pp. 3879–3891). DOI: 10.18653/v1/2023.findings-emnlp.254.

[C10] **Li, Z.**, Zhang, C., Jin, Y., Cang, X., Puntambekar, S., & Passonneau, R. J. (2023, June). Learning when to defer to humans for short answer grading. In *International Conference on Artificial Intelligence in Education* (pp. 414–425). Cham: Springer Nature Switzerland. (AIED 2023)

[C11] Lloyd, S., Beckman, M. D., Pearl, D. K., Passonneau, R. J., **Li, Z.**, & Wang, Z. (2022, September). Foundations for AI-Assisted Formative Assessment Feedback for Short-Answer Tasks in Large-Enrollment Classes. In *Proceedings of the 11th International Conference on Teaching Statistics (ICOTS 11)*. DOI: 10.52041/iase.icots11.T3C3.

[C12] **Li, Z.**, Tomar, Y., & Passonneau, R. J. (2021, November). A semantic feature-wise transformation relation network for automatic short answer grading. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (pp. 6030–6040). (EMNLP 2021)

[C13] **Li, Z.**, Feng, Y., Xu, J., Guo, J., Lan, Y., & Cheng, X. (2019, July). Teaching machines to extract main content for machine reading comprehension. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 33, No. 1, pp. 9973–9974).

[C14] **Li, Z.**, Xu, J., Lan, Y., Guo, J., Feng, Y., & Cheng, X. (2018, September). Hierarchical answer selection framework for multi-passage machine reading comprehension. In *China Conference on Information Retrieval* (pp. 93–104). Cham: Springer International Publishing. (**Best Paper Nominee**)

[C15] Guo, T., Xu, J., Yan, X., Hou, J., Li, P., **Li, Z.**, … & Cheng, X. (2016, October). Ease the process of machine learning with dataflow. In *Proceedings of the 25th ACM International Conference on Information and Knowledge Management (CIKM 2016)* (pp. 2437–2440). (**Best Demo Paper Nominee**)

[C16] Wang, Y.; **Li, Z.**; Liu, J.; He, Z.; Huang, Y.; Li, D. (2014). *Word Vector Modeling for Sentiment Analysis of Product Reviews*. NLPCC 2014.

Journal Articles

[J1] Akemoğlu, Y., **Li, Z.**, Zheng, Q., Roberts, E. B., Singh, S. P., & Xiong, J. (2026). Parent's AI Coach (PaiCoach): A Single-Case Experimental Study of an AI-Supported Parent Coaching System for Autistic Children. *Journal of Autism and Developmental Disorders*. Published online September 11, 2026. DOI: 10.1007/s10803-026-07519-6.

[J2] Xiao, F., **Li, Z.**, Lin, J., Zou, X., Yang, D., Zou, E. W., & Xiong, J. (2025). Leveraging an LLM-enhanced bilingual conversational agent for EFL children's dialogic reading: Insights from children, parents, and educators. *Computers and Education: Artificial Intelligence*, 100484.

[J3] Xiao, F., Zou, E. W., Lin, J., **Li, Z.**, & Yang, D. (2025). Parent-led vs. AI-guided dialogic reading: Evidence from a randomized controlled trial in children's e-book context. *British Journal of Educational Technology*, 56(5), 1784–1813.

[J4] Passonneau, R., Koenig, K., **Li, Z.**, & Soddano, J. (2023). The ideal versus the real deal in assessment of physics lab report writing. *European Journal of Applied Sciences*, 11(2).

Edited Books

[B1] Akemoğlu, Y., Xiong, J., **Li, Z.**, & Zheng, Q. (Eds.). *Human-Centered AI for Special Education – Design, Assessment, and Intervention* (provisional title). Springer. **Under contract; in preparation.**

Datasets & Preprints

[R1] **Li, Z.**, & Passonneau, R. J. (2024). Joint Training for Selective Prediction. *arXiv preprint* arXiv:2410.24029.

[R2] *Dataset.* **Li, Z.**; Lloyd, S.; Beckman, M. D.; Passonneau, R. J. (2023). *ISTUDIO: A Statistical Automatic Response Assessment Dataset*. Penn State Data Commons.

[R3] *Dataset.* **Li, Z.**; Atil, B.; Koenig, K.; Passonneau, R. J. (2022). *Reliable Rubric-Based Assessment of Physics Lab Reports: Data for Machine Learning*. Penn State Data Commons.